

School of Engineering,
Cochin University of Science And Technology
B. Tech. Degree Second Semester First Internal Examination April 2024
23-200-0203A ENGINEERING MECHANICS (CE A Batch, CE B Batch and ME B Batch)
(2023 scheme)

Duration: 2 hours

Maximum Marks: 30

Course Outcomes: On completion of the course, a student will be able to

1. Explain principles and theorems related to rigid body mechanics.
2. Identify the components of a system of forces acting on the rigid body.
3. Apply the conditions of equilibrium to various practical problems involving different force system.
4. Choose appropriate theorems, principles or formulae to solve problems of mechanics.
5. Solve problems involving rigid bodies, applying the properties of distributed areas and masses.

Part A

(Answer ALL questions)

		(5 x 2 = 10 marks)			
Q. No	QUESTIONS	Marks	BL	CO	PO
I. (a)	Explain resolution of a force?	(2)	L2	[CO1]	1,2
(b)	A man who has mass of 72 kg is standing on a board inclined 20° with horizontal. Find the component of man's weight perpendicular and parallel to plane of board	(2)	L1	[CO2]	1,2
(c)	List the types of force systems	(2)	L4	[CO1]	1,2
(d)	Explain simple harmonic motion with suitable examples	(2)	L2	[CO1]	1,2
(e)	The rectilinear motion of a particle is defined by the displacement time equation $x = t^4 - 3t^3 + 2t^2 - 8$. find velocity and acceleration at time $t = 2$ sec.	(2)	L3	[CO4]	1,2

Part B

(Answer any TWO questions)

		(2 x 10 = 20 marks)			
II. (a)	Three forces acting on a particle 'O' as shown in figure1. Determine the value of 'F' such that resultant of these forces is horizontal. Find the magnitude and direction of fourth force which when acting along with given 3 forces will keep 'O' in equilibrium.	(5)	L5	[CO3]	1,2

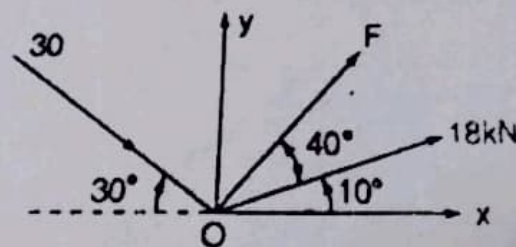


Fig 1

(b)	A roller of weight $W = 1000$ N rests on a smooth inclined plane and is kept from rolling down by a string AC as shown in figure2. Evaluate the tension, S in the string and the reaction R_B at the point of contact B.	(5)	L5	[CO3,4]	1,2
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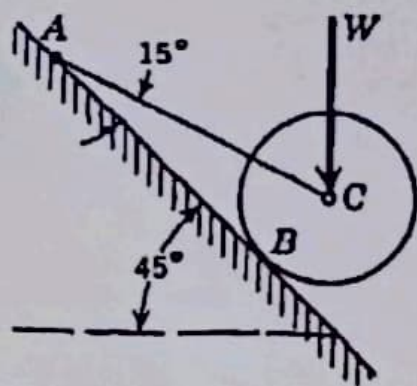


Fig 2

OR

- III. (a) A weight of 800 N is supported by two chains as shown in figure 3. Determine the tension in each chain.

(4) L5 [CO3,4] 1,2

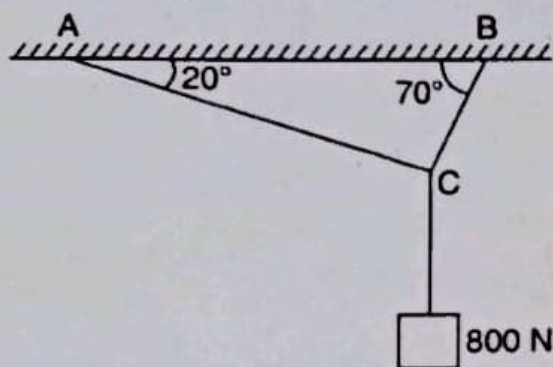


Fig 3

- (b) A chord supported at A and B carries a load of 10 kN at D and a load of W at C as shown in figure 4. Find the value of W so that CD remains horizontal

(6) L3 [CO3,4] 1,2

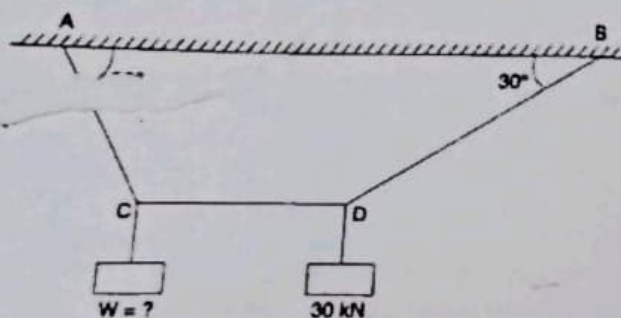


Fig 4

- IV (a) Determine the velocity and acceleration of a piston of heat engine when it is at a distance of 75 cm from the center and crank rotates at 120rpm. Take stroke length as 200cm.

(5) L5 [CO4] 1,2

- (b) A mass of 400g shown in Figure 5 is connected to a light spring whose spring constant is 5kN/m. It is free to oscillate on a horizontal frictionless track. If the mass is displaced 10cm from equilibrium and released from rest, find

(5) L3 [CO4] 1,2

- period of motion
- maximum velocity of mass
- maximum acceleration of mass
- equations for displacement, velocity and acceleration as

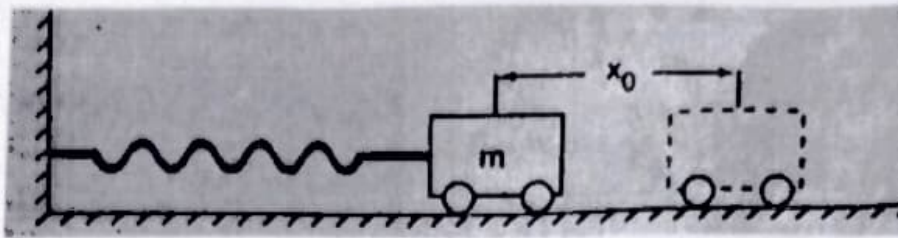


Fig 5

OR

- IV A car travelling at 40 Kmph sights a distant signal at 150m and comes uniformly to rest at the signal. It remains at rest for 20sec. As allowed by the signal, it uniformly accelerates and attains 40Kmph in 250m. Calculate the time lost due to the presence of signal for the entire journey. (10) L5 [CO4] 1,2